Exp No: 17

Linear Regression.

Program:

```
import numpy as np import pandas as pd
df=pd.read_csv('Salary_data.csv') df df.info()
df.dropna(inplace=True) df.info() df.describe()
features=df.iloc[:,[0]].values label=df.iloc[:,[1]].values
from sklearn.model_selection import train_test_split
x_train,x_test,y_train,y_test=train_test_split(features,l
abel, test_size=0.2,random_state=23) from
sklearn.linear_model import LinearRegression
model=LinearRegression() model.fit(x_train,y_train)
model.score(x_train,y_train)
model.score(x_test,y_test) model.coef_
model.intercept_ import pickle
pickle.dump(model,open('SalaryPred.model','wb'))
model=pickle.load(open('SalaryPred.model','rb'))
yr_of_exp=float(input("Enter Years of Experience: "))
yr_of_exp_NP=np.array([[yr_of_exp]])
Salary=model.predict(yr_of_exp_NP) print("Estimated
Salary for {} years of experience is {}: "
.format(yr_of_exp,Salary))
```

Output:

	YearsExperience	Salary
0	1.1	39343
1	1.3	46205
2	1.5	37731
3	2.0	43525
4	2.2	39891

```
<class
'pandas.core.frame.DataFrame'>
RangeIndex: 30 entries, 0 to
29 Data columns (total 2
columns):
 #  Column          Non-Null Count
Dtype
---  ------          --------------  -----
 0  YearsExperience 30 non-null
float64  1  Salary          30 non-null
int64 dtypes: float64(1), int64(1)
memory usage: 612.0 bytes
```

Output:

	YearsExperience	Salary
count	30.000000	30.000000
mean	5.313333	76003.000000
std	2.837888	27414.429785
min	1.100000	37731.000000
25%	3.200000	56720.750000
50%	4.700000	65237.000000
75%	7.700000	100544.750000
max	10.500000	122391.000000

LinearRegression()

0.960318254743891

0.918417084921423

array([[9281.30847068078]])

array([27166.73682891278])

Output:

Enter Years of Experience: 44 Estimated Salary for 44.0 years of experience is [[435544.30953887]]: